

Technical Architecture & Detailed Solution Proposal

High-Altitude Anti-Drone System (DRDO PS 26050)

1. Reframed Problem Statement: The High-Altitude Challenge

The objective outlined in DRDO PS 26050 extends far beyond developing a generic detect and track drone system. High altitude border environments introduce a compound matrix of physical extremes: severe sub zero temperatures, drastically reduced atmospheric pressure, high-velocity winds, and mechanical vibration.

Commercial and standard military off the shelf (MOTS) anti-drone systems fail in these environments because they operate blindly. They assume a nominal operating environment and rely solely on passive, heavy ruggedization (like thick enclosures) to survive. However, physical survival does not equate to operational accuracy. As temperatures drop, mechanical lubricants thicken, cables stiffen, and battery internal resistance spikes. As air pressure drops, standard convection cooling for edge-computing electronics fails. As wind increases, optical tracking gimbals are blown off target.

The true problem to solve is maintaining performance stability and micro-tracking accuracy across volatile altitudes and climates. We must transition from a system that merely "survives" the environment to one that actively measures and compensates for it.

2. Step-by-Step Solution Architecture

Phase 1: The Baseline Multi-Modal Detection Problem

Problem: Small Unmanned Aerial Vehicles (UAVs) present exceptionally low Radar Cross Sections (RCS) and often utilize autonomous, radio silent flight paths, making single-sensor detection highly unreliable.

Solution: We implement a Multi-Modal Detection Suite integrated via an Edge AI compute unit (such as the NVIDIA Jetson Orin Nano class). This suite includes:

- **FMCW/mmWave Radar:** Provides continuous range and velocity tracking, functioning day and night. While clutter can be an issue, radar remains resilient against optical fog.
- **High-Resolution EO Camera:** Delivers vital visual identification and payload classification, though vulnerable to lens fogging and nighttime limitations.
- **Thermal/LWIR Imaging:** Detects heat signatures. Crucially, the extreme cold ambient temperatures of high altitudes act as an advantage here, significantly improving the thermal delta (contrast) between a drone's motors and the background sky.
- **Passive RF Sensing:** A receive-only antenna array to detect control and telemetry link

frequencies, acting as an early warning system before physical line-of-sight is established.

Phase 2: The Environmental Blindness Problem

Problem: The sensors listed in Phase 1, along with their mechanical gimbals, suffer severe performance degradation under high-altitude stress. If the system does not know it is degrading, it will report false negatives or inaccurate tracking vectors.

Solution: We introduce an independent Environmental State & System Health Estimation Layer. We deploy secondary industrial-grade diagnostic sensors throughout the chassis:

- **Environmental Polling:** Using BME280/BMP390 sensors to measure ambient temperature, humidity, and barometric pressure. The pressure reading acts as a proxy for altitude and air density.
- **Vibration & Orientation:** Industrial IMUs (like the BMI088) monitor the acceleration noise floor to detect mechanical fatigue and platform shaking induced by wind.
- **Internal Subsystem Health:** We monitor the battery using INA219/INA226 current/voltage sensors to watch for cold induced voltage sag. Thermistors monitor the motor drivers, and high precision magnetic encoders measure the exact physical response of the mechanical actuators against the commands they were given.

Phase 3: The Mechanical Stiffening & Wind Disturbance Problem

Problem: At -30°C, external silicone cables stiffen and bearing grease becomes highly viscous. Simultaneously, high altitude winds exert a physical disturbance torque on the camera gimbal, pushing it off the target.

Solution: We deploy intelligent Precision Pointing & Stabilization control loops governed by an STM32-class industrial microcontroller. The system calculates a "**torque residual**" the difference between the commanded tracking angle and the actual angle achieved (read via magnetic encoders). If the system commands a 10-degree pan, but the motor lags, the software instantly identifies this as environmental stiffening (e.g., Cold Temp → Cable Stiffens → Gimbal Torque Increases → Encoder Tracking Lag Increases). Instead of failing, the system automatically adapts its PID (Proportional-Integral-Derivative) control gains and injects a feed-forward compensation term to push through the wind and mechanical resistance, restoring ~0.1–0.5° pointing accuracy.

Phase 4: The Conflicting Sensor Problem

Problem: If a sudden snowstorm rolls in, the optical (EO) camera's feed will become noisy and useless. A standard system's tracking algorithm might still try to average the EO camera's data with the radar data, corrupting the final tracking output.

Solution: We implement Dynamic Sensor Fusion using an **Extended Kalman Filter (EKF) and Weighted Bayesian Fusion**. Explanation and Mechanism of Kalman Filter

is after Detailed Challenge Accommodation Matrix

Because the system is aware of the environment (Phase 2), it actively alters the trust weights of its sensors. If the optical dust sensor detects occlusion, or image variance drops due to fog, the software actively down-weights the EO camera's confidence score to near zero, shifting 100% of the tracking reliance to the radar and thermal arrays. This ensures the output trajectory is never corrupted by a blinded sensor.

3. The Complete Solution: Environmental Aware Adaptive Compensation

By integrating the solutions to the four problems above, we achieve our core innovation. Our proposed architecture is an Environmental Aware Adaptive Compensation Layer. The operational flow operates as a continuous, closed-loop pipeline:

Environmental Sensors → Environmental State Estimation → System Health Estimation → Degradation Prediction → Adaptive Compensation (PID gain scheduling / Thermal heating) → Tracking Correction → Unified Confidence Score → Operator Dashboard.

The system is not just hardware; it is a self-diagnosing platform. Operators interacting with the React based tactical dashboard do not just see a camera feed; they see real-time degradation curves, per-modality confidence bars, and the active compensation metrics proving the system is keeping itself locked onto the target.

4. Detailed Challenge Accommodation Matrix

Hardware Integration Framework

The system is designed as a ruggedized, highly integrated sensor node. The components are grouped into four main functional blocks:

- **Housing & Environmental Control:** A Bud Industries IP65 NEMA enclosure forms the main chassis, stabilized on a Neewer Heavy-Duty Tripod. Internal pressure is regulated by an M12 waterproof vent. Thermal management relies on a Stego PTC Heater 028, a heat spreader, and custom-cut silicone thermal barrier sheets to segment the internal climate. All internal mounts are 3D-printed from durable PETG.
- **Power Architecture:** Energy is stored in stable LiFePO4 cells and meticulously managed by a Daly Smart BMS (4S 100A). Power is then routed through a dedicated Power Distribution Board (PDB) to provide clean, isolated 12V rails to the compute and mechanical systems.
- **Compute & Sensor Suite:** The NVIDIA Jetson Orin NX acts as the central fusion controller, communicating via SPI, I2C, and UART. It synchronizes a Texas Instruments IWR6843 mmWave radar for drone detection and a FLIR Lepton 3.5 LWIR thermal camera for visual confirmation, managed through an I2C multiplexer.

- **Motion & Telemetry:** Physical tracking is handled by azimuth and elevation gimbals driven by SimpleFOC Mini Brushless Drivers. Secure, long-range telemetry is maintained by a Microchip RN2903 LoRa module forming an encrypted radio mesh.

Environmental Factor	Physical Effect & Risk	Observable Parameter	System Accommodation / Compensation
Severe Cold (Low Temp)	Increases internal battery resistance (capacity drop) and increases lubricant viscosity in gimbal bearings.	Battery voltage sag; increased motor current draw to achieve movement.	Software initiates power-aware gain scheduling. Hardware utilizes active resistive heating pads selectively on the LiFePO4 battery bay (isolating heat from optics to prevent condensation).
Low Atmospheric Pressure	Drastically reduces the effectiveness of convection cooling, threatening edge-compute (Jetson/Radar) with thermal throttling.	Internal case and CPU/GPU thermistor temperatures rise despite low ambient external temperatures.	Physical accommodation via zoned thermal design: intensive passive heat spreading to the external enclosure walls, separating the vented compute bay from the insulated battery bay.
High Wind & Gusts	Applies disturbance torque on the pan-tilt mechanism, pulling the camera off the target drone.	Position error (delta between commanded angle and encoder feedback).	Implementation of feed-forward disturbance-observer control in the STM32 MCU, instantly applying counter-torque.
Vibration	Creates sensor noise in the IMU, corrupting tracking stability, and causes long-term mechanical/solder fatigue.	Spikes in the accelerometer noise floor.	Mechanical silicone isolation mounts for the gimbal, paired with software-level Extended Kalman Filter (EKF) bias correction to filter out high-frequency noise.
Dust / Snow / Fog	Optical occlusion completely blinding the EO camera, leading to false negatives.	Drastic drop in image contrast and noise ratio in OpenCV processing.	Fusion re-weighting: The software dynamically ignores the EO feed, shifts reliance to Radar/Thermal, and triggers a cleaning cycle alert on the dashboard.

Conclusion: By moving the burden of high-altitude operation from pure physical ruggedization into intelligent, continuous software level compensation, this proposed

architecture provides DRDO with a highly resilient, verifiable, and constantly optimized anti drone tracking system.

Environmental / Operational Challenge	Integrated Component	Engineering Solution
Moisture, Dust, and Wind	Bud Industries NEMA Enclosure	The IP65 rating ensures the delicate fusion controller and sensors are completely isolated from water jets, high humidity, and dust ingress common at high altitudes.
Atmospheric Pressure Fluctuations	M12 Waterproof Vent	As altitude changes, internal enclosure pressure can blow out seals or damage sensors. The vent allows the system to "breathe" and equalize pressure without compromising the IP65 moisture barrier.
Extreme Cold / Freezing Temperatures	Stego PTC Heater 028 & Heat Spreader	Thin air and high-altitude cause freezing that can stop mechanical parts and kill batteries. The PTC heater provides a self-regulating thermal baseline, distributed evenly by the heat spreader to keep the internal ambient temperature within operational limits.
Localized Hotspots & Thermal Gradients	Silicone Thermal Barrier Sheet	High-power compute boards generate heat that can distort LWIR thermal camera readings. The silicone pad physically separates heat-generating boards from sensitive payloads, preventing interference.
Battery Thermal Runaway & Over-discharge	Daly Smart BMS & LiFePO4 Cells	High UV exposure and extreme temperature swings make standard lithium batteries volatile. LiFePO4 chemistry provides inherent thermal stability, while the BMS actively prevents over-discharge and monitors cell health.
Voltage Sags from Mechanical Actuators	Power Distribution	Gimbal motors draw massive power spikes during rapid tracking maneuvers. The PDB isolates the 12V rails,

Environmental / Operational Challenge	Integrated Component	Engineering Solution
	Board (PDB)	preventing actuator noise and voltage drops from resetting the NVIDIA Jetson controller.
Mechanical Vibration & Tracking Instability	SimpleFOC Mini Drivers & PETG Mounts	High-altitude winds cause micro-vibrations that ruin radar and thermal imaging. PETG mounts provide rigid, impact-resistant structural support, while SimpleFOC drivers use precise PID control loops to counter wind buffeting and maintain a lock on targets.
Data Bottlenecks & Sensor Desync	NVIDIA Jetson Orin NX & I2C Multiplexer	Processing radar point clouds and thermal video simultaneously can crash standard microcontrollers. The Orin NX provides high-bandwidth data processing, while the multiplexer ensures the I2C bus doesn't jam when both sensors poll simultaneously.
Interception & Loss of Connectivity	Microchip RN2903 LoRa Module	Standard Wi-Fi or cellular fails at remote altitudes. LoRa provides a low-power, long-range radio mesh, encrypted to ensure drone tracking telemetry cannot be intercepted or jammed by countermeasures.

1. Gaussian Distribution

Starting with a scalar Gaussian distribution for a random variable X :

$$f_X(x) = ce^{-\frac{(x-\mu)^2}{2\sigma^2}} \equiv \mathcal{N}(\mu, \sigma^2)$$

where μ is the mean, σ is the standard deviation, and $c = \frac{1}{\sqrt{2\pi}\sigma}$ is the normalization constant required to ensure the area under the probability density function equals 1.

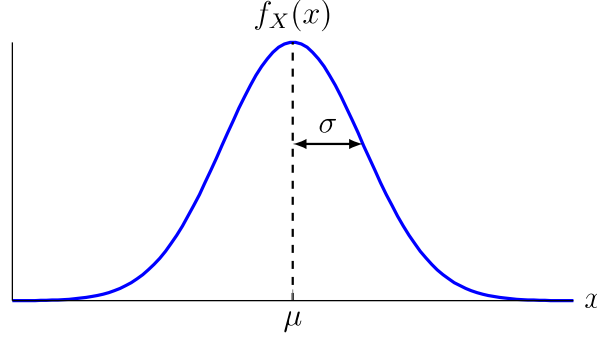


Figure 1: Probability density function of a 1D Gaussian distribution centered at μ .

Extending this for scalar variables (X, Y) to find the joint Gaussian distribution, we define the state vector Z :

$$Z = \begin{bmatrix} X \\ Y \end{bmatrix}$$

Assuming zero-mean ($\mu = 0$) for simplicity, let Σ be the covariance matrix defined as the expected value of Z multiplied by its transpose:

$$\begin{aligned} \Sigma &= E[ZZ^T] \\ &= E\left(\begin{bmatrix} X \\ Y \end{bmatrix} \begin{bmatrix} X & Y \end{bmatrix}\right) \\ &= E\left(\begin{bmatrix} X^2 & XY \\ YX & Y^2 \end{bmatrix}\right) \\ &= \begin{bmatrix} E[X^2] & E[XY] \\ E[XY] & E[Y^2] \end{bmatrix} \end{aligned}$$

Here, $E[X^2]$ represents the variance $\text{var}(X) = \sigma_X^2$, and $E[XY]$ represents the covariance $\text{cov}(X, Y) = \sigma_{XY}$. Thus:

$$\Sigma = \begin{bmatrix} \sigma_X^2 & \sigma_{XY} \\ \sigma_{XY} & \sigma_Y^2 \end{bmatrix}$$

To solve for the exponent term $Z^T \Sigma^{-1} Z$ in the joint distribution, we first evaluate the determinant of the covariance matrix:

$$|\Sigma| = \sigma_X^2 \sigma_Y^2 - \sigma_{XY}^2$$

Next, we expand the quadratic form matrix multiplication:

$$\begin{aligned} Z^T \Sigma^{-1} Z &= \begin{bmatrix} X & Y \end{bmatrix} \frac{1}{|\Sigma|} \begin{bmatrix} \sigma_Y^2 & -\sigma_{XY} \\ -\sigma_{XY} & \sigma_X^2 \end{bmatrix} \begin{bmatrix} X \\ Y \end{bmatrix} \\ &= \frac{1}{|\Sigma|} (X^2 \sigma_Y^2 - 2XY \sigma_{XY} + Y^2 \sigma_X^2) \end{aligned}$$

We treat Y as a known constant (a measurement) and explicitly complete the square for the X terms. First, isolate the terms containing X and factor out σ_Y^2 :

$$X^2 \sigma_Y^2 - 2XY \sigma_{XY} = \sigma_Y^2 \left(X^2 - 2X \left(Y \frac{\sigma_{XY}}{\sigma_Y^2} \right) \right)$$

Now, complete the square inside the parenthesis by adding and subtracting $\left(Y \frac{\sigma_{XY}}{\sigma_Y^2} \right)^2$:

$$\begin{aligned} &= \sigma_Y^2 \left[\left(X - Y \frac{\sigma_{XY}}{\sigma_Y^2} \right)^2 - Y^2 \frac{\sigma_{XY}^2}{\sigma_Y^4} \right] \\ &= \sigma_Y^2 \left(X - Y \frac{\sigma_{XY}}{\sigma_Y^2} \right)^2 - Y^2 \frac{\sigma_{XY}^2}{\sigma_Y^2} \end{aligned}$$

Substitute this back into the full exponent term $Z^T \Sigma^{-1} Z$ along with the leftover $Y^2 \sigma_X^2$:

$$Z^T \Sigma^{-1} Z = \frac{1}{|\Sigma|} \left[\sigma_Y^2 \left(X - Y \frac{\sigma_{XY}}{\sigma_Y^2} \right)^2 - Y^2 \frac{\sigma_{XY}^2}{\sigma_Y^2} + Y^2 \sigma_X^2 \right]$$

Combine the isolated Y terms:

$$\begin{aligned} -Y^2 \frac{\sigma_{XY}^2}{\sigma_Y^2} + Y^2 \sigma_X^2 &= Y^2 \left(\frac{\sigma_X^2 \sigma_Y^2 - \sigma_{XY}^2}{\sigma_Y^2} \right) \\ &= Y^2 \frac{|\Sigma|}{\sigma_Y^2} \end{aligned}$$

Distributing the $\frac{1}{|\Sigma|}$ across both sections, let this final exponent term be T :

$$T = \frac{\sigma_Y^2}{|\Sigma|} \left(X - Y \frac{\sigma_{XY}}{\sigma_Y^2} \right)^2 + \frac{Y^2}{\sigma_Y^2}$$

The joint PDF is given by $f_{X,Y}(X,Y) = c \cdot \exp(-\frac{T}{2})$. Because T is a sum of two distinct terms, the exponential explicitly splits into a product of two functions:

$$\begin{aligned} f_{X,Y}(X,Y) &= c \cdot \exp \left(-\frac{1}{2} \left[\frac{\sigma_Y^2}{|\Sigma|} \left(X - Y \frac{\sigma_{XY}}{\sigma_Y^2} \right)^2 + \frac{Y^2}{\sigma_Y^2} \right] \right) \\ &= \exp \left(-\frac{1}{2} \frac{\sigma_Y^2}{|\Sigma|} \left(X - Y \frac{\sigma_{XY}}{\sigma_Y^2} \right)^2 \right) \cdot \exp \left(-\frac{1}{2} \frac{Y^2}{\sigma_Y^2} \right) \\ &\equiv f_{X|Y}(X|Y) \cdot f_Y(Y) \end{aligned}$$

Initially, we assumed $\mu_X = 0$ and $\mu_Y = 0$. Restoring the general means by replacing $X \rightarrow X - \mu_X$ and $Y \rightarrow Y - \mu_Y$, the conditional mean becomes:

$$\mu_{X|Y} = \mu_X + \frac{\sigma_{XY}}{\sigma_Y^2} (Y - \mu_Y)$$

We have now officially derived the joint Gaussian PDF for (X,Y) from scratch.

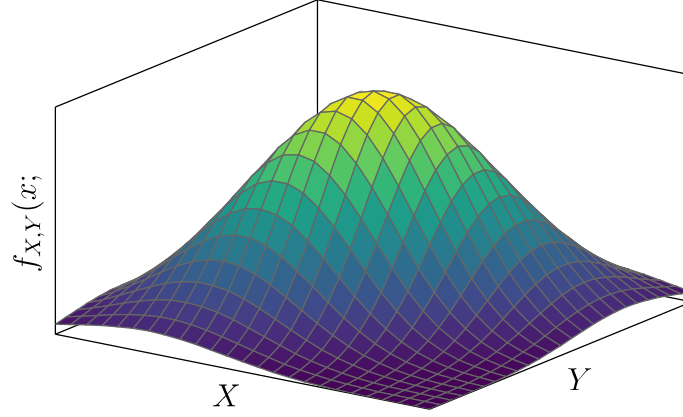


Figure 2: Surface plot of a 2D Joint Gaussian distribution. The shape, spread, and orientation of this 3D probability surface are entirely governed by the covariance matrix Σ .

2. Vectors and Schur Complement

Now assume X and Y are column vector quantities instead of scalars: $X \in \mathbb{R}^{p \times 1}$ and $Y \in \mathbb{R}^{q \times 1}$. The block covariance matrices are formally defined as:

$$\begin{aligned}\Sigma_{XX} &\equiv \text{cov}(X, X) \in \mathbb{R}^{p \times p} \\ \Sigma_{XY} &\equiv \text{cov}(X, Y) \in \mathbb{R}^{p \times q} \\ \Sigma_{YX} &\equiv \text{cov}(Y, X) \in \mathbb{R}^{q \times p} \\ \Sigma_{YY} &\equiv \text{cov}(Y, Y) \in \mathbb{R}^{q \times q}\end{aligned}$$

To find Σ^{-1} , we use block matrix inversion. Let M represent the block covariance matrix partitioned as:

$$M = \begin{bmatrix} E & F \\ G & H \end{bmatrix}$$

where E, F, G, H are sub-matrices corresponding to the dimensions of the covariance blocks. We introduce arbitrary vectors U and V to solve the linear system $MZ = [U \ V]^T$:

$$\begin{bmatrix} E & F \\ G & H \end{bmatrix} \begin{bmatrix} X \\ Y \end{bmatrix} = \begin{bmatrix} U \\ V \end{bmatrix}$$

Which yields the system of linear equations:

$$\begin{aligned}EX + FY &= U \\ GX + HY &= V\end{aligned}$$

Assuming block H is invertible, we isolate Y from the second equation:

$$\begin{aligned}HY &= V - GX \\ \implies Y &= H^{-1}(V - GX)\end{aligned}$$

Substituting Y into the first equation:

$$\begin{aligned}EX + F(H^{-1}(V - GX)) &= U \\ EX - FH^{-1}GX &= U - FH^{-1}V \\ (E - FH^{-1}G)X &= U - FH^{-1}V\end{aligned}$$

Let $S = E - FH^{-1}G$. This term S is known as the **Schur Complement** of block H in matrix M . We can now solve for X :

$$X = S^{-1}U - S^{-1}FH^{-1}V$$

Substituting X back to solve fully for Y :

$$\begin{aligned} Y &= H^{-1}V - H^{-1}G(S^{-1}U - S^{-1}FH^{-1}V) \\ &= -H^{-1}GS^{-1}U + (H^{-1} + H^{-1}GS^{-1}FH^{-1})V \end{aligned}$$

Mapping these coefficients back to the matrix inverse yields:

$$M^{-1} = \begin{bmatrix} S^{-1} & -S^{-1}FH^{-1} \\ -H^{-1}GS^{-1} & H^{-1} + H^{-1}GS^{-1}FH^{-1} \end{bmatrix}$$

Now compute the exponent term $Z^T \Sigma^{-1} Z$ for the vector case. Substituting M^{-1} :

$$\begin{aligned} Z^T \Sigma^{-1} Z &= \begin{bmatrix} X^T & Y^T \end{bmatrix} \begin{bmatrix} S^{-1} & -S^{-1}FH^{-1} \\ -H^{-1}GS^{-1} & H^{-1} + H^{-1}GS^{-1}FH^{-1} \end{bmatrix} \begin{bmatrix} X \\ Y \end{bmatrix} \\ &= X^T S^{-1} X - X^T S^{-1} F H^{-1} Y \\ &\quad - Y^T H^{-1} G S^{-1} X + Y^T (H^{-1} + H^{-1} G S^{-1} F H^{-1}) Y \end{aligned}$$

Since covariance matrices are strictly symmetric, we know $E = E^T$, $H = H^T$, and $F = G^T$.

$$\begin{aligned} S &= E - FH^{-1}F^T \\ \implies S &= S^T \end{aligned}$$

Treating Y as a known measurement vector, we complete the square on the general quadratic matrix form. First, we establish the identity:

$$X^T E X + 2h^T E X = (X + h)^T E (X + h) - h^T E h$$

Let the remainder term be $K = -h^T E h$.

Comparing terms with our derived $Z^T \Sigma^{-1} Z$ expression, we identify:

$$E = S^{-1} \quad \text{and} \quad 2h^T S^{-1} = -2Y^T H^{-1} G S^{-1}$$

Solving for h :

$$\begin{aligned} h &= -G^T H^{-1} Y \\ &= -F H^{-1} Y \end{aligned}$$

Now, explicitly calculate the remainder term K :

$$\begin{aligned} K &= -h^T E h \\ &= -(-Y^T H^{-1} G) S^{-1} (-F H^{-1} Y) \\ &= -Y^T H^{-1} G S^{-1} F H^{-1} Y \end{aligned}$$

Substitute $(X + h)^T E(X + h) + K$ back into $Z^T \Sigma^{-1} Z$, combining it with the leftover Y terms from our initial expansion:

$$\begin{aligned} Z^T \Sigma^{-1} Z &= (X - FH^{-1}Y)^T S^{-1} (X - FH^{-1}Y) \\ &\quad - Y^T H^{-1} G S^{-1} F H^{-1} Y \\ &\quad + Y^T (H^{-1} + H^{-1} G S^{-1} F H^{-1}) Y \end{aligned}$$

Notice that distributing the last term gives $Y^T H^{-1} Y + Y^T H^{-1} G S^{-1} F H^{-1} Y$. The second part perfectly cancels the remainder K .

$$Z^T \Sigma^{-1} Z = (X - FH^{-1}Y)^T S^{-1} (X - FH^{-1}Y) + Y^T H^{-1} Y$$

Finally, substitute this completed square back into the joint vector Gaussian distribution. The exponential explicitly factors into two distinct probability densities:

$$\begin{aligned} \exp\left(-\frac{1}{2} Z^T \Sigma^{-1} Z\right) &= \exp\left(-\frac{1}{2} [(X - FH^{-1}Y)^T S^{-1} (X - FH^{-1}Y) + Y^T H^{-1} Y]\right) \\ &= \exp\left(-\frac{1}{2} (X - FH^{-1}Y)^T S^{-1} (X - FH^{-1}Y)\right) \cdot \exp\left(-\frac{1}{2} Y^T H^{-1} Y\right) \\ &\equiv f_{X|Y}(X|Y) \cdot f_Y(Y) \end{aligned}$$

The conditional distribution is thus mathematically proven to be Gaussian:

$$X|Y = y \sim \mathcal{N}(\mu_X + \Sigma_{XY} \Sigma_{YY}^{-1} (Y - \mu_Y), \Sigma_{XX} - \Sigma_{XY} \Sigma_{YY}^{-1} \Sigma_{YX})$$

3. Minimum Mean Square Error (MMSE) Optimality

Let the projection of X onto the vector space of Y be defined as a linear transformation BY . The orthogonality principle states the error $(X - BY)$ is orthogonal ($\perp$) to Y :

$$E[(X - BY)Y^T] = 0$$

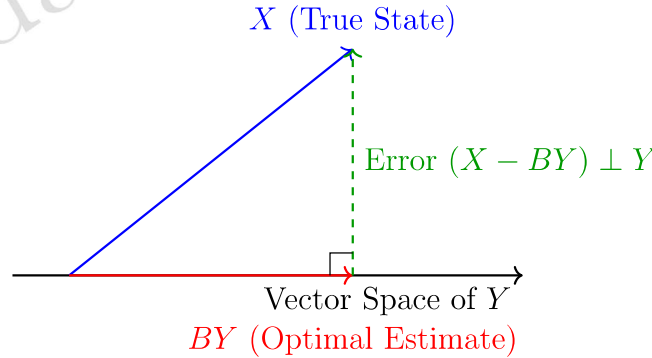


Figure 3: The Orthogonality Principle: The optimal MMSE linear estimate BY projects X such that the error is perfectly orthogonal to the observation space.

Distributing the expectation:

$$\begin{aligned} E[XY^T] - E[BYY^T] &= 0 \\ E[XY^T] - BE[YY^T] &= 0 \\ \implies B\Sigma_{YY} &= \Sigma_{XY} \\ \implies B &= \Sigma_{XY} \Sigma_{YY}^{-1} \end{aligned}$$

The mean in the joint Gaussian PDF of term $X|Y$ is exactly the projection of X onto the vector space of Y . We have shown that the minimum MSE for a linear estimator occurs when $B = \Sigma_{XY}\Sigma_{YY}^{-1}$.

To prove global MMSE optimality, let $G(Y)$ be any arbitrary estimator function. Let the optimal estimate be the conditional expectation $E[X|Y]$.

$$(X - G(Y))^2 = ((X - E[X|Y]) + (E[X|Y] - G(Y)))^2$$

Define the optimal estimation error γ and the estimator deviation λ :

$$\begin{aligned}\gamma &= X - E[X|Y] \\ \lambda &= E[X|Y] - G(Y)\end{aligned}$$

Substituting these into the squared error equation:

$$E[(X - G(Y))^2] = E[\gamma^2] + E[\lambda^2] + 2E[\gamma\lambda]$$

By the law of total expectation, we can evaluate the cross-term:

$$\begin{aligned}E[\gamma\lambda] &= E[E[\gamma\lambda|Y]] \\ &= E[\lambda E[\gamma|Y]]\end{aligned}$$

We now expand the inner conditional expectation $E[\gamma|Y]$:

$$\begin{aligned}E[\gamma|Y] &= E[(X - E[X|Y])|Y] \\ &= E[X|Y] - E[E[X|Y]|Y] \\ &= E[X|Y] - E[X|Y] \\ &= 0\end{aligned}$$

Since the inner expectation is zero, the cross-term vanishes:

$$E[\gamma\lambda] = 0$$

Returning to the main error expectation equation:

$$\begin{aligned}E[(X - G(Y))^2] &= E[\gamma^2] + E[\lambda^2] \\ &\geq E[\gamma^2]\end{aligned}$$

Thus, we have rigorously proved that $E[X|Y]$ is the best possible estimate, as any other estimator $G(Y)$ will add a positive squared error λ^2 , meaning it will always be greater than or equal to $E[\gamma^2]$.

4. Forming the Basic Model for the Kalman Filter

Before assembling the Kalman filter equations, we must explicitly state the core assumptions governing the physical system and the sensors:

- **Linearity:** The state transitions and measurement observations are governed purely by linear mathematical relationships.

- **Gaussian Noise:** Both the process noise within the system and the measurement noise from the sensors are drawn from zero-mean Gaussian (Normal) distributions.
- **Independence:** The process noise, the measurement noise, and the initial state are mutually independent from one another.

Under these assumptions, let x_k be the hidden state vector to be estimated at discrete time step k . The physical process model is defined as:

$$x_k = Ax_{k-1} + w_k \quad (1)$$

where A is the state transition matrix and w_k is the Gaussian process noise.

The measurement model is defined as:

$$y_k = Cx_k + n_k \quad (2)$$

where y_k is the measurement vector, C is the observation matrix, and n_k is the Gaussian measurement noise. Let Q and R be the covariance matrices for w_k and n_k , respectively. A, C, Q, R comprise the known system parameters.

Let $\hat{x}_k^-$ denote the *a priori* estimate (the prediction before observing y_k), and P_k^- be the *a priori* error covariance matrix.

$$\begin{aligned} \hat{x}_k^- &\equiv E[x_k | y_{1:k-1}] \\ &= E[Ax_{k-1} + w_k | y_{1:k-1}] \end{aligned}$$

Since the process noise w_k is zero-mean:

$$\hat{x}_k^- = A\hat{x}_{k-1} \quad (3)$$

Let the *a priori* estimation error be e_k^- :

$$\begin{aligned} e_k^- &\equiv x_k - \hat{x}_k^- \\ &= A(x_{k-1} - \hat{x}_{k-1}) + w_k \\ &= Ae_{k-1} + w_k \end{aligned}$$

The error covariance is calculated as $P_k^- = E[e_k^- (e_k^-)^T]$:

$$\begin{aligned} P_k^- &= E[(Ae_{k-1} + w_k)(w_k^T + e_{k-1}^T A^T)] \\ &= AE[e_{k-1} e_{k-1}^T] A^T + E[w_k w_k^T] \\ P_k^- &= AP_{k-1} A^T + Q \end{aligned} \quad (4)$$

Equations (3) and (4) form the **Predict** step.

Let $\hat{x}_k$ and P_k be the *a posteriori* (updated) estimate and error covariance after the actual measurement y_k is observed. Using our joint Gaussian derivations from Section 2:

$$\begin{aligned} \Sigma_{XY} &= \text{cov}(x_k, Cx_k + n_k) \\ &= P_k^- C^T \end{aligned}$$

$$\begin{aligned} \Sigma_{YY} &= \text{cov}(Cx_k + n_k) \\ &= CP_k^- C^T + R \end{aligned}$$

Defining the Kalman gain K_k as the term $\Sigma_{XY}\Sigma_Y^{-1}$ derived during our MMSE proof:

$$K_k = P_k^- C^T (C P_k^- C^T + R)^{-1} \quad (5)$$

Substituting this into our optimal mean and covariance equations yields the **Update** step:

$$\hat{x}_k = \hat{x}_k^- + K_k(y_k - C\hat{x}_k^-) \quad (6)$$

$$P_k = P_k^- - K_k C P_k^- \quad (7)$$

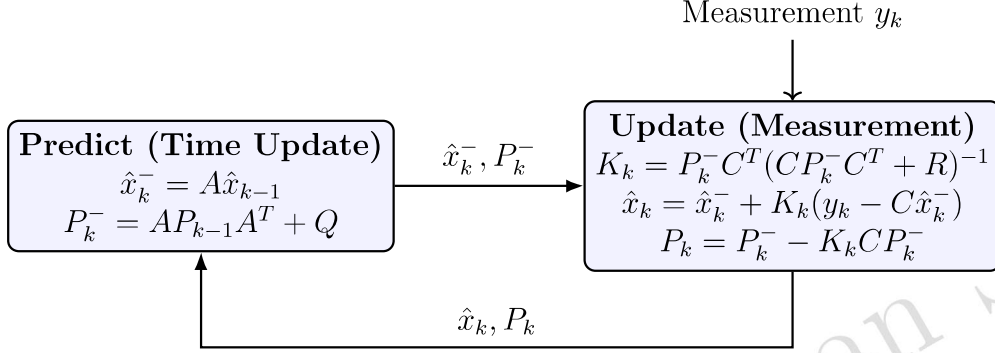


Figure 4: The recursive nature of the Kalman Filter: Predictions are made based on system dynamics, then updated/corrected using sensor measurements.

Conclusion & Acknowledgments

In conclusion, the sequence above represents a complete, from-scratch mathematical derivation of the standard linear, Gaussian-noise Kalman filter. By progressing from the foundational properties of joint Gaussian distributions and block matrix inversion to the formal proof of Minimum Mean Square Error (MMSE) optimality, we have rigorously established both the predict and update stages of the algorithm.

This derivation and understanding were heavily inspired by the foundational teachings of Prof. Gilbert Strang (Linear Algebra) and Prof. John Tsitsiklis (MIT OCW 6.041 Probabilistic Systems Analysis and Applied Probability).

Prerequisite Research for PS-50: High-Altitude Performance Optimization and Robust Design of an Anti-Drone System

1. Research Context

PS-50 focuses on developing an **anti-drone system capable of reliable detection, identification, tracking, and threat assessment under high-altitude and changing environmental conditions**. At increased altitude, factors such as reduced atmospheric pressure and air density, low temperature, strong or variable winds, changing visibility, sensor noise, communication limitations, and power constraints can affect both the target drone and the anti-drone system.

Therefore, the prerequisite research focused on understanding **how drones are built, how they fly, how they communicate and navigate, what signatures they produce, and how these characteristics may change under different operating conditions**.

2. Drone Anatomy and Architecture

The basic architecture of a UAV was studied through its major components, including the **frame, motors, propellers, Electronic Speed Controllers (ESCs), Power Distribution Board (PDB), flight controller, receiver and transmitter, battery, GPS, telemetry system, sensors, FPV camera, and Video Transmitter (VTX)**.

Different physical configurations were also examined, including **X-frames, H-frames, hybrid H-frames, Z-frames, unibody and split-arm designs**, as well as quadcopters, hexacopters, and octacopters.

Understanding this architecture is essential for PS-50 because every component contributes to the drone's physical, electromagnetic, visual, thermal, and behavioral characteristics. These characteristics can potentially be exploited by an external detection system for **drone identification and classification**.

3. Types and Applications of Drones

The research covered different UAV configurations, including **multirotors, fixed-wing, hybrid, and VTOL drones**.

Drones were also studied according to their applications, including **surveillance, scientific research, rescue, cinematography, racing/FPV, delivery, agriculture, and combat-oriented applications.**

This classification is important because different UAV types exhibit different **flight dynamics, speeds, maneuverability, altitude capabilities, physical structures, and detectable signatures.** Consequently, an anti-drone system should not rely on a single motion or appearance model.

4. Propulsion, Flight Mechanics and Aerodynamics

The functioning of drone propulsion systems was studied through **motors, propellers, ESCs, throttle, thrust, and motor-direction configurations.**

The fundamental flight parameters of **roll, pitch, yaw, and throttle** were examined to understand how a UAV controls its orientation, altitude, and movement.

Basic aerodynamics was also studied, including the behavior of fixed-wing UAVs and control surfaces such as **ailerons, elevators, and rudders**, along with concepts such as leading and trailing edges and bank angle.

These concepts directly support PS-50 because **wind and changes in atmospheric density at high altitude can alter the flight behavior and trajectory of UAVs.** An anti-drone tracking system must therefore account for environmental disturbances instead of assuming that the target always follows an ideal trajectory.

5. Flight Controller, IMU and Sensors

The **Flight Controller (FC)** was studied as the central control unit responsible for interpreting sensor information and generating appropriate motor commands.

The research covered the **Inertial Measurement Unit (IMU), gyroscope, accelerometer, barometer, magnetometer, GPS and other sensors.**

The gyroscope and accelerometer provide information about angular motion and acceleration, while the barometer can assist with altitude estimation, the

magnetometer contributes to heading estimation, and GPS provides positioning and timing information.

This is highly relevant to high-altitude operation because **temperature, pressure, wind, GPS conditions, and sensor noise can influence state estimation and consequently the flight behavior of the UAV.**

6. PID Control and Flight Stability

The research examined **PID control**, consisting of proportional, integral, and derivative components.

PID-based control allows a flight controller to compare the desired state of the drone with its measured state and continuously apply corrections. This enables stable flight despite disturbances.

For PS-50, understanding PID control is important because environmental disturbances such as strong winds can cause the UAV to continuously make corrective movements. These corrections affect its **acceleration, velocity, heading, and trajectory**, which are precisely the parameters an external tracking system attempts to estimate.

Thus, understanding flight-control behavior helps distinguish between **normal environmentally induced movement and genuinely anomalous movement.**

7. Communication Systems and UART

The communication architecture of UAVs was studied through **transmitter-receiver systems, TX/RX communication, serial communication and UART (Universal Asynchronous Receiver-Transmitter).**

These communication pathways may carry control commands, telemetry, configuration information, or other data between different components and between the UAV and its operator.

This provides the foundation for understanding **RF-based detection and signal tracking**, which can form one of the sensing layers of an anti-drone system.

However, communication reliability can vary with distance, interference, antenna orientation, environmental conditions, and autonomous operation.

Therefore, communication-based detection should ideally complement rather than replace other sensing modalities.

8. FPV Systems, Cameras and Video Transmission

FPV drones and their **cameras, VTX systems, receivers, and video-transmission architecture** were studied, including the concept of **latency**, or the delay between real-world movement and its appearance in the received video.

This research is relevant to the optical component of an anti-drone system. Cameras can provide valuable information for **visual detection, identification and classification**, but their performance may be affected by lighting, distance, atmospheric conditions, fog, clouds and visibility.

This reinforces the need for multiple sensing modalities, particularly under high-altitude conditions.

9. Telemetry, Navigation and Ground-Control Systems

The research covered **GPS, telemetry, waypoints, heading, altitude, GPS time, battery status, flight mode, artificial horizon, aircraft attitude, bank angle and cross-track error**.

Software platforms including **Betaflight, INAV, ArduPilot and Mission Planner** were studied to understand how UAVs are configured, navigated, monitored and controlled.

Mission Planner, in particular, provides insight into how parameters such as **GPS position, altitude, heading, telemetry status, flight mode and aircraft orientation** can be monitored through a ground-control environment.

Understanding these parameters provides the conceptual foundation for designing an external system capable of estimating similar information from independent sensors.

10. High-Altitude Environmental Challenges

The most important aspect of the prerequisite research is understanding how **high-altitude conditions influence both UAV operation and anti-drone sensing.**

As altitude increases, atmospheric density and pressure decrease, while temperature and wind conditions can vary significantly. These changes can affect propulsion efficiency, battery performance, aerodynamic behavior, sensor measurements, communication reliability and target visibility.

At the same time, the anti-drone system may experience **sensor noise, reduced optical visibility, altered signal characteristics, increased tracking uncertainty and increased false detections.**

Therefore, the anti-drone system cannot be designed solely around nominal ground-level conditions.

11. Radar, RF, Optical and Infrared Detection

The research introduced three major categories of external drone detection:

Radar-Based Detection

Radar can provide information about the presence, position and movement of aerial targets and can contribute to long-range detection and tracking.

RF/Signal-Based Detection

RF monitoring can potentially identify electromagnetic activity associated with drone communication systems, providing another source of information for detection and classification.

Optical and Infrared Detection

Visible-light cameras can provide visual identification, while infrared sensing can provide complementary information under different lighting and visibility conditions.

The key research conclusion is that **each sensing modality has limitations.** Consequently, combining radar, RF, optical and infrared information can provide a more robust basis for UAV detection.

12. Sensor Fusion and Adaptive Detection

All the preceding research ultimately leads to the concept of **multi-sensor fusion**.

Instead of treating every sensor as equally reliable at all times, the anti-drone system can estimate the current operating environment and determine the reliability of each sensing modality.

Conceptually, the system follows:

Environmental Conditions → Sensor Reliability → Sensor Fusion → Detection → Classification → Tracking → Threat Assessment

For example, if visibility deteriorates, optical confidence may decrease while other sensing modalities continue to provide useful information. Similarly, if wind increases target-motion uncertainty, the tracking system should adapt its prediction model rather than interpreting every deviation as an anomaly.

This directly addresses the central requirement of PS-50.

13. Edge AI and Intelligent Processing

The research also explored **Edge AI and local intelligent processing**.

Instead of depending entirely on remote computation, sensor data can be processed locally to enable faster decision-making and reduce dependence on external connectivity.

For PS-50, Edge AI can potentially be used to estimate **environmental conditions, sensor confidence, target classification, trajectory uncertainty and threat probability**.

This allows the sensing and tracking pipeline to dynamically respond to changing conditions rather than operating with fixed parameters.

14. Drone Swarms and Coordinated Behavior

The study of **drone swarms, swarm intelligence, communication topology and bio-inspired algorithms** introduced the challenge of multiple UAVs operating cooperatively.

A swarm requires the anti-drone system to perform **multi-target detection and tracking**, maintain the identity of individual targets, and potentially identify correlated or coordinated movement.

This is an extension of the robustness problem because the system must remain reliable not only when environmental conditions change, but also when the number and behavior of aerial targets become more complex.

15. Threat Assessment and High-Risk UAVs

The study of **loitering munitions and other potentially harmful UAV applications** highlighted the importance of moving beyond simple drone detection.

A robust anti-drone system should ultimately be able to distinguish between ordinary UAV activity and behavior that may indicate a potential threat.

However, environmental conditions must be considered during this assessment. A trajectory deviation caused by strong wind should not automatically be interpreted as malicious behavior.

Therefore, threat assessment should incorporate **trajectory, altitude, velocity, direction, behavior, target classification and environmental context**.

17. Final Research Insight

The prerequisite study establishes the foundation for PS-50 by connecting the **internal functioning of UAVs** with the **external challenges of detecting and tracking them at high altitude**.

Understanding motors, propellers, ESCs, flight controllers, IMUs, GPS, telemetry, communication systems, FPV architecture, flight modes, aerodynamics and autonomous navigation explains how drones behave. Understanding radar, RF tracking, optical and infrared sensing explains how that behavior can be observed externally.

Most importantly, studying high-altitude environmental effects demonstrates why a fixed detection and tracking pipeline may not remain reliable under changing conditions.

Hence, the proposed direction is an **adaptive, multi-sensor and AI-assisted anti-drone architecture** in which the system continuously estimates its operating conditions, evaluates sensor reliability, fuses available information, and dynamically adjusts its detection and tracking strategy.

The fundamental objective is therefore not merely to detect a drone, but to maintain reliable detection, identification and tracking despite changes in altitude, weather, visibility, sensor quality, communication conditions and target behavior. This forms the core technical motivation behind PS-50.